# Sambhav Balki

**Codechef** | **Codeforces** | **Leetcode** | **GFG**
Department of Computer Science and Engineering
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| Degree/Board | Institution | Year | CGPA/% |
|---|---|---|---|
| B.Tech, CSE | National Institute of Technology Raipur | 2024-28 | 9.08/10.0 |
| HSC – XII | Janata Junior College, Chandrapur | 2023 | 83% |
| SSC – X | Chanda Public School, Chandrapur | 2021 | 95.6% |

## Technical Skills

**Languages**: C/C++, Java, Python, JavaScript , SQL
**Backend**: Node.js, Express.js
**Database**: MongoDB
**Frontend**: HTML5, CSS3, Tailwind CSS, EJS
**Utilities**: Git, GitHub
**Soft Skills**: Leadership, Quizzing, Designing
**Core Subjects**: OOP, Computer Networks, Operating Systems , DBMS
**Operating System**: Windows
**Data Science / ML**: NumPy, Pandas, Matplotlib

## Projects

**PlanForm.AI**

*An AI system converting floor plan images into structured spatial graphs and interactive 3D models* 2026

- Achieved **90%+ accuracy** in detecting structural features using a **Computer Vision pipeline (OpenCV)**.
- Converted floor plans into **spatial graphs** enabling structured reasoning and **room-level inference**.
- Built interactive **3D visualizations** using **Three.js**, improving plan interpretability by **3x**.
- Designed a **material recommendation engine** optimizing cost and structural strength across **100+ materials**.
- Integrated **Groq LLM** to generate contextual explanations of architectural decisions.
- **Tech Stack:** React, Node.js, FastAPI, OpenCV, NumPy, Three.js, TailwindCSS, Groq API

**Pravah** *(In Development)*

*A role-based project management platform for student collaboration, deadlines, and tracking* 2026

- Architected a role-based workflow for **Students, Teachers, and Admins** to streamline project collaboration.
- Implemented rigid **role-based access control (RBAC)** and secure session handling across user actions.
- Built project submission, deadline tracking, and real-time management features using **MERN architecture**.
- Developed a **RAG-powered knowledge assistant** for conversational search over legacy project reports.
- **Tech Stack:** React, Node.js, Express.js, MongoDB, Mongoose

**Brotel**

*Full-stack hostel and homestay booking platform with property management and authentication* 2026

- Developed a **full-stack application** with complete CRUD functionality for flexible property listings.
- Implemented secure user authentication and route protection using stateful **Express Sessions**.
- Built a robust **RESTful MVC backend** using **Express.js** and **MongoDB** with comprehensive error handling.
- Implemented **image upload workflows** using **Multer** and cloud/local asset persistence layouts.
- Designed responsive, reusable **EJS views** integrated with TailwindCSS for unified rendering.
- **Tech Stack:** Node.js, Express.js, MongoDB, Mongoose, EJS, HTML5, CSS3, TailwindCSS, Multer

## Achievements

- Codeforces Round 1073 (Div. 2) **Rank: 9949**
- Codeforces Round 1068 (Div. 2) **Rank: 8864**
- Codeforces **Max Rating: 1065**
- CodeChef Starters 237 (Rated) **Global Rank: 350**
- CodeChef **Max Rating: 1460**
- Leetcode **Max Rating: 1752**
- Prompt-Thon IIIT Naya Raipur **Stood among Top 5 teams out of 100+ participants**
- CodeStrike 2026 **Secured position among Top 10 teams out of 1500+ participants**

## Positions of Responsibility

**Event Manager**

*Entrepreneurship Cell, NIT Raipur* 2025 – Present

- Managed and coordinated entrepreneurship-related events and competitions with 100+ participants.
- Led end-to-end event execution including planning, logistics, and on-ground coordination.
- Collaborated with core team members to ensure smooth event operations and engagement.